



Laboratory work №5

Databases

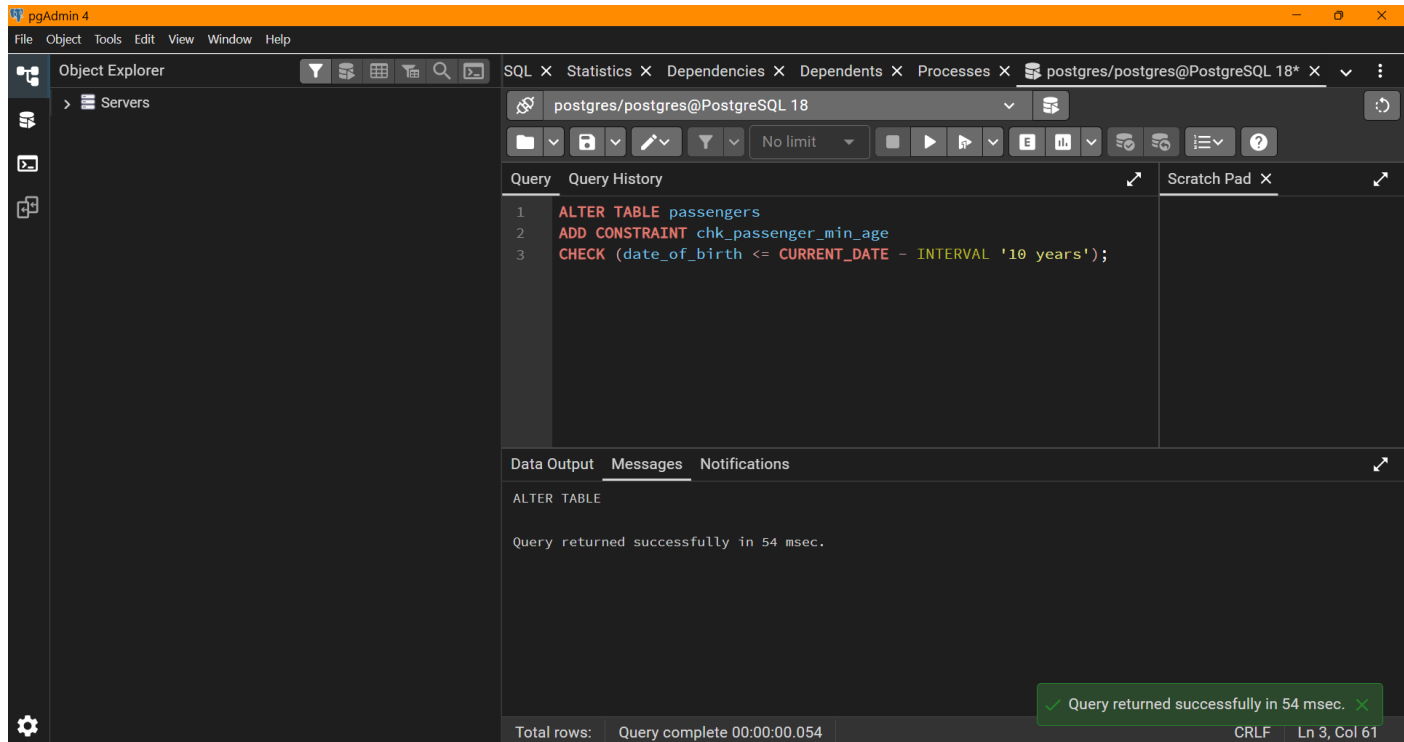
Tolepказы Galamat

Tuesday, 11:00

Almaty 2026

Tasks:

1. Add a CHECK constraint to passenger table to provide that passengers must be at least 10 years old.



The screenshot shows the pgAdmin 4 interface. The SQL editor contains the following query:

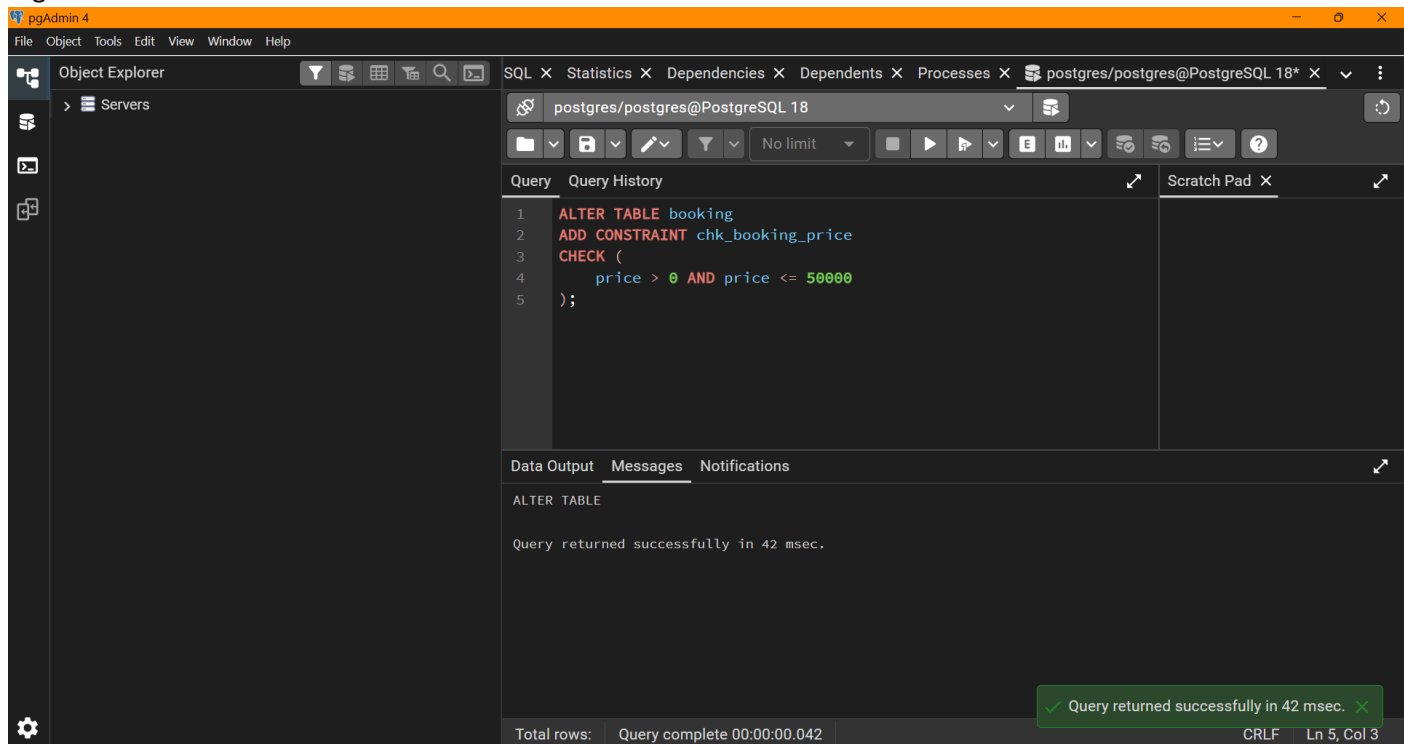
```
1 ALTER TABLE passengers
2 ADD CONSTRAINT chk_passenger_min_age
3 CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '10 years');
```

The Messages pane shows the following output:

```
ALTER TABLE
Query returned successfully in 54 msec.
```

A green status bar at the bottom right indicates: "Query returned successfully in 54 msec." The status bar also shows "Total rows: Query complete 00:00:00.054" and "CRLF Ln 3, Col 61".

2. Add a CHECK constraint to accept values in booking price not more than 50000tg and less than 0tg.



The screenshot shows the pgAdmin 4 interface. The SQL editor contains the following query:

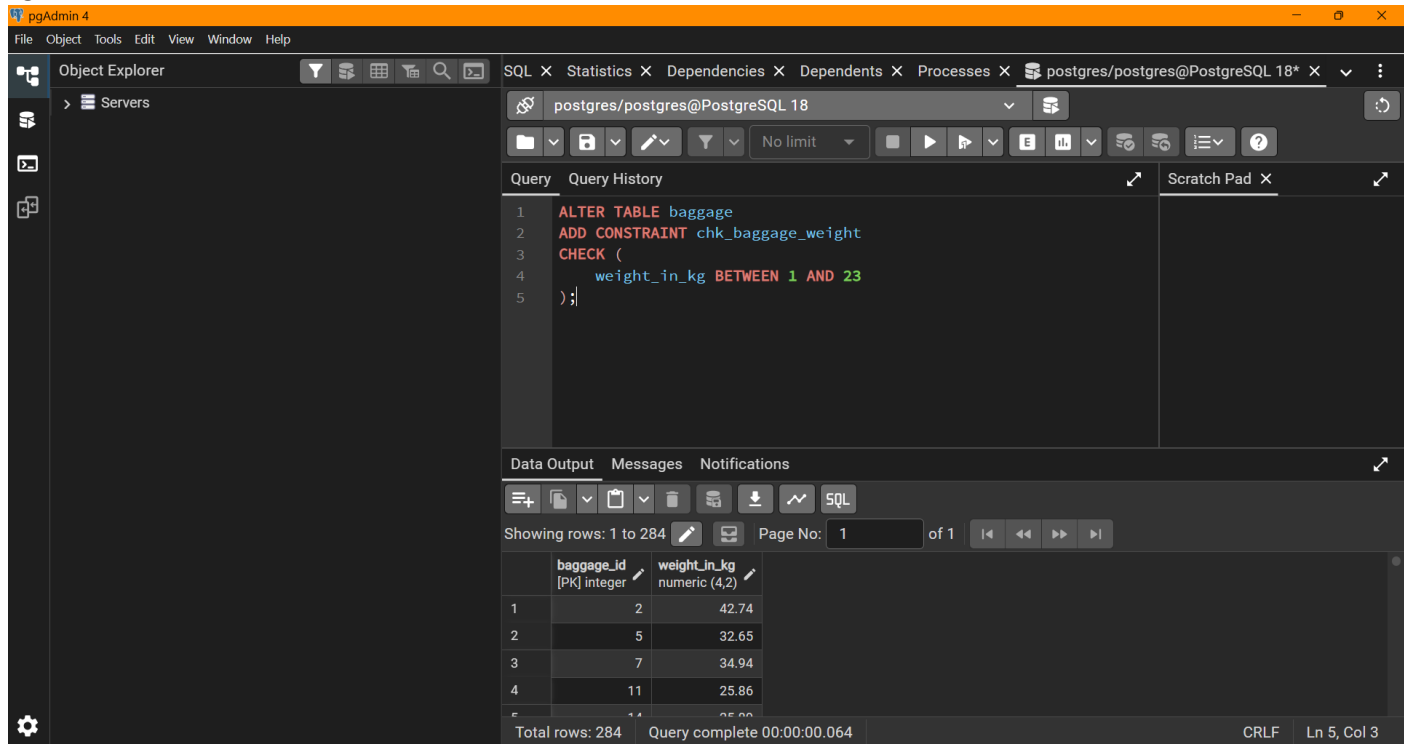
```
1 ALTER TABLE booking
2 ADD CONSTRAINT chk_booking_price
3 CHECK (
4 price > 0 AND price <= 50000
5 );
```

The Messages pane shows the following output:

```
ALTER TABLE
Query returned successfully in 42 msec.
```

A green status bar at the bottom right indicates: "Query returned successfully in 42 msec." The status bar also shows "Total rows: Query complete 00:00:00.042" and "CRLF Ln 5, Col 3".

3. Add a CHECK constraint to accept the luggage weight between 1 and 23 kg.



The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following code:

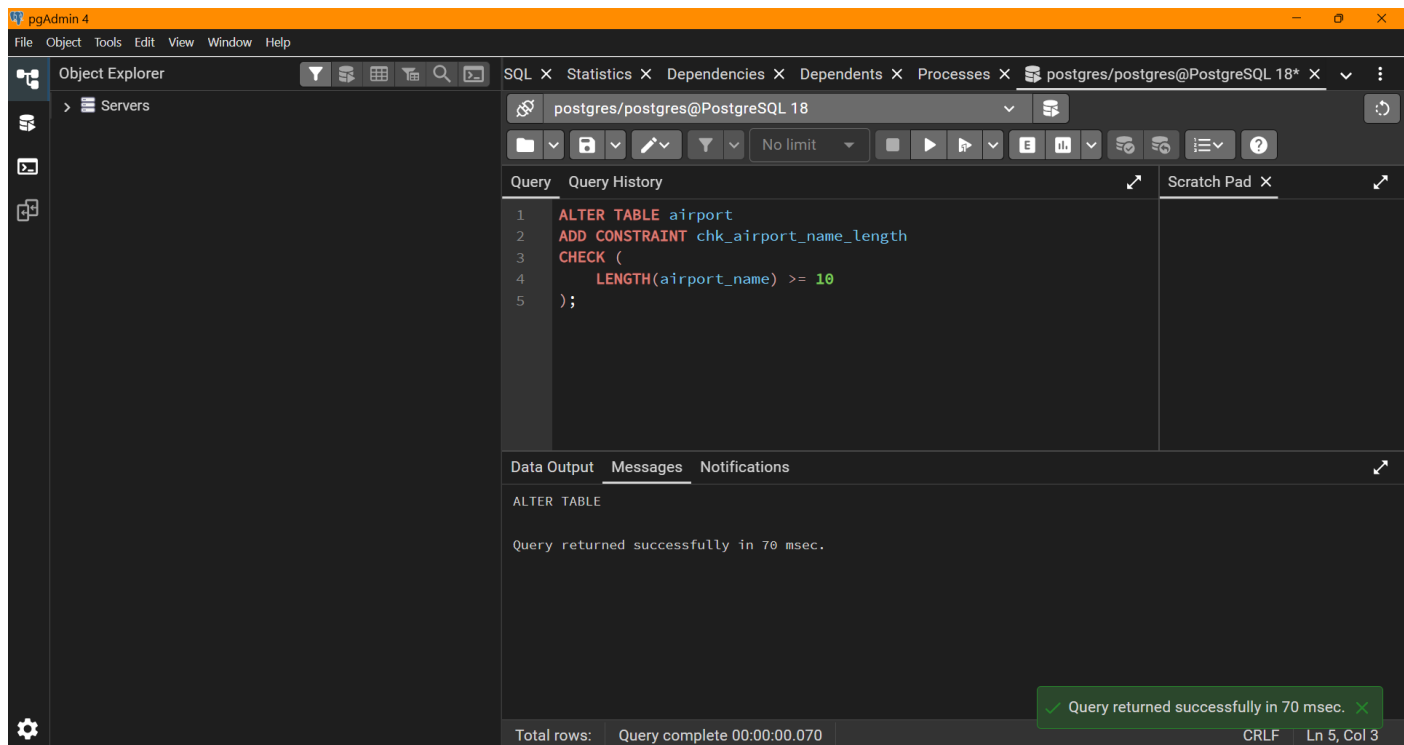
```
1 ALTER TABLE baggage
2 ADD CONSTRAINT chk_baggage_weight
3 CHECK (
4     weight_in_kg BETWEEN 1 AND 23
5 );
```

The Data Output tab shows the results of the query, displaying a table with two columns: baggage_id [PK] integer and weight_in_kg numeric (4,2). The table contains four rows of data:

baggage_id [PK] integer	weight_in_kg numeric (4,2)
1	42.74
2	32.65
3	34.94
4	25.86

The status bar at the bottom indicates "Total rows: 284" and "Query complete 00:00:00.064".

4. Add a CHECK constraint to ensure that all values in airport_name must have at least 10 characters.



The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following code:

```
1 ALTER TABLE airport
2 ADD CONSTRAINT chk_airport_name_length
3 CHECK (
4     LENGTH(airport_name) >= 10
5 );
```

The Data Output tab shows the results of the query, displaying the text "ALTER TABLE" and "Query returned successfully in 70 msec.".

A green notification box at the bottom right states: "Query returned successfully in 70 msec.".

The status bar at the bottom indicates "Total rows: 0" and "Query complete 00:00:00.070".

5. Add UNIQUE constraint to some columns in each table in database.

The screenshot shows the pgAdmin 4 interface with the SQL editor open. The query being executed is:

```
1 ALTER TABLE passengers
2 ADD CONSTRAINT uq_passenger_passport
3 UNIQUE (passport_number);
4
5 ALTER TABLE airline
6 ADD CONSTRAINT uq_airline_code
7 UNIQUE (airline_code);
8
9
```

The Data Output pane shows the message: "Query returned successfully in 44 msec." A green status bar at the bottom indicates "Query returned successfully in 44 msec." The status bar also shows "Total rows: Query complete 00:00:00.044" and "CRLF Ln 9, Col 1".

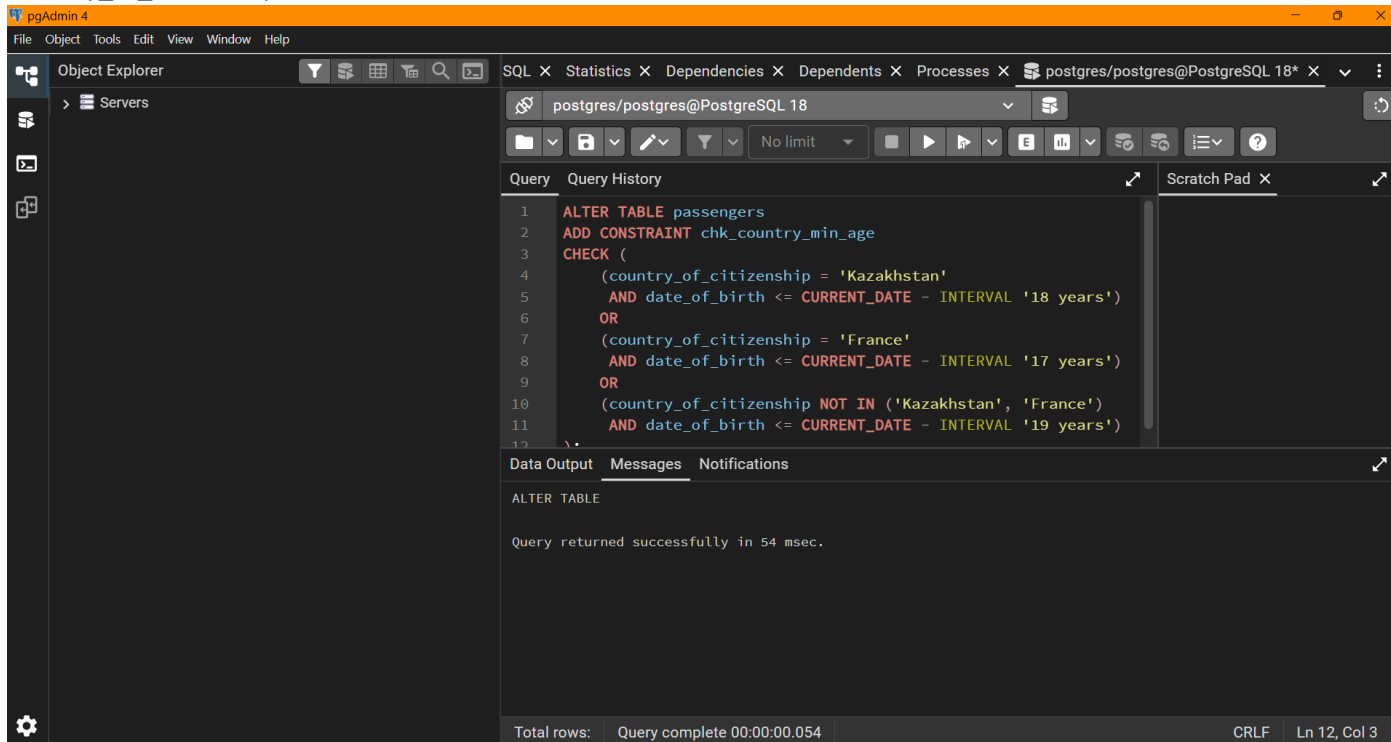
6. Add a CHECK constraint to ensure that male passengers must be at least 18 years old and female passengers must be 19 years old.

The screenshot shows the pgAdmin 4 interface with the SQL editor open. The query being executed is:

```
1 ALTER TABLE passengers
2 ADD CONSTRAINT chk_gender_min_age
3 CHECK (
4     (gender = 'Male'
5      AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
6     OR
7     (gender = 'Female'
8      AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
9 ) not valid;
```

The Data Output pane shows the message: "Query returned successfully in 54 msec." A green status bar at the bottom indicates "Query returned successfully in 54 msec." The status bar also shows "Total rows: Query complete 00:00:00.054" and "CRLF Ln 9, Col 12".

7. Add a CHECK constraint to add rule as follow (use column country_of_citizenship):



The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following SQL code:

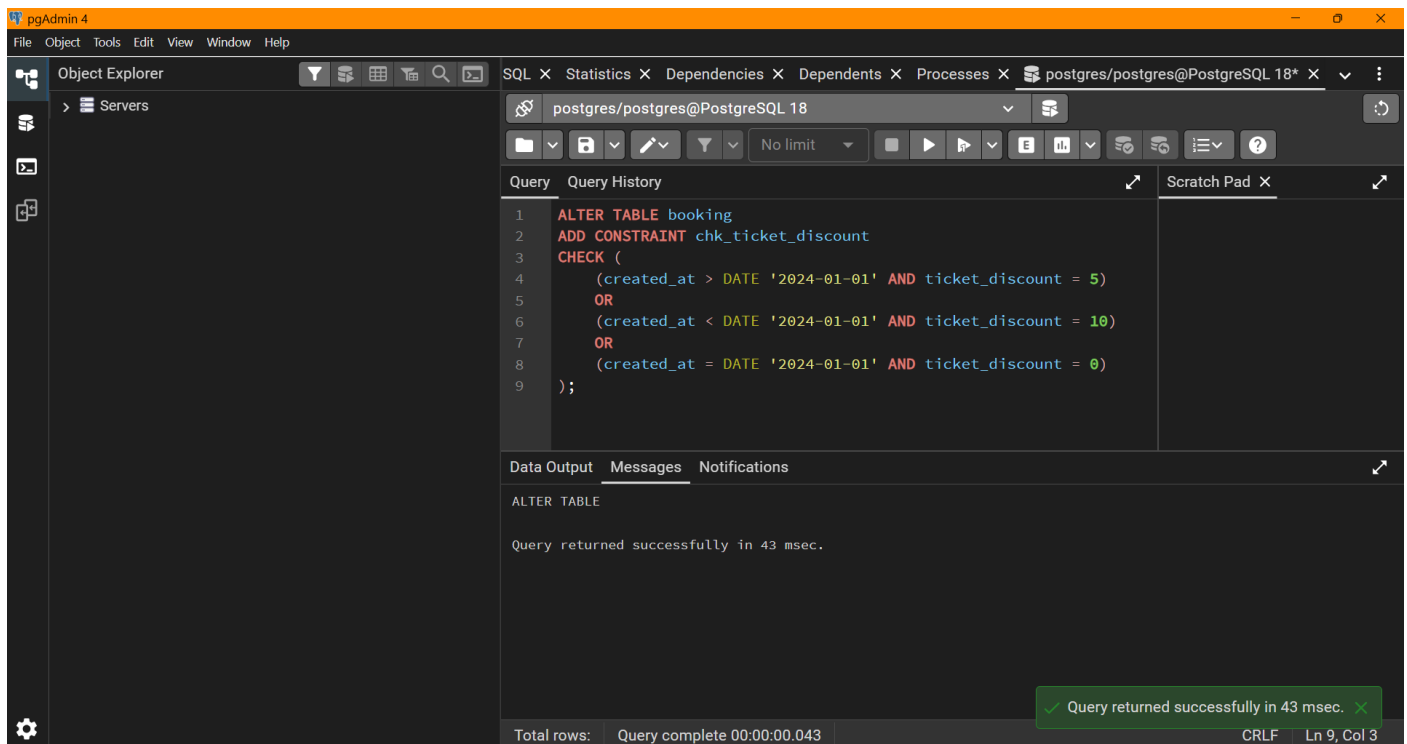
```
1 ALTER TABLE passengers
2 ADD CONSTRAINT chk_country_min_age
3 CHECK (
4     (country_of_citizenship = 'Kazakhstan'
5      AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
6     OR
7     (country_of_citizenship = 'France'
8      AND date_of_birth <= CURRENT_DATE - INTERVAL '17 years')
9     OR
10    (country_of_citizenship NOT IN ('Kazakhstan', 'France')
11     AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
12 );
```

The Messages pane shows the following output:

```
ALTER TABLE
Query returned successfully in 54 msec.
```

The status bar at the bottom indicates: Total rows: Query complete 00:00:00.054 CRLF Ln 12, Col 3

- Passengers from Kazakhstan must be at least 18 years old.
 - Passengers from France must be at least 17 years old.
 - Passengers from other countries must be at least 19 years old.
8. Add a ticket_discount column to table booking and a CHECK constraint to apply some discount based on ticket price and created time:
- the constraint applies a 5% discount for tickets created after 2024-01-01,
- and 10% discount for tickets created before 2024-01-01-
- 01.



The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following SQL code:

```
1 ALTER TABLE booking
2 ADD CONSTRAINT chk_ticket_discount
3 CHECK (
4     (created_at > DATE '2024-01-01' AND ticket_discount = 5)
5     OR
6     (created_at < DATE '2024-01-01' AND ticket_discount = 10)
7     OR
8     (created_at = DATE '2024-01-01' AND ticket_discount = 0)
9 );
```

The Messages pane shows the following output:

```
ALTER TABLE
Query returned successfully in 43 msec.
```

The status bar at the bottom indicates: Total rows: Query complete 00:00:00.043 CRLF Ln 9, Col 3

A green notification bar at the bottom right states: ✓ Query returned successfully in 43 msec. ✕

9. Modify the booking table so that the booking status column has a default value of 'Confirmed'. Then insert a new booking without specifying booking

status.

The screenshot shows the pgAdmin 4 interface. The SQL editor contains the following query:

```
1 INSERT INTO booking (  
2     booking_id,  
3     passenger_id,  
4     booking_platform,  
5     created_at,  
6     update_at,  
7     price,  
8     ticket_discount  
9 )  
10  
11 VALUES (  
12     1001,  
13     1  
14 )
```

The Data Output tab shows the results of the query:

	booking_id [PK] integer	passenger_id integer	booking_platform character varying (50)	created_at date	update_at date	status character varying (50)	price numeric
1	1001	1	Online	2026-10-07	2026-10-07	Confirmed	1
2		1	Frami Inc	2023-06-06	2023-08-01	Male	
3		2	Ullrich-Mueller	2024-01-03	2024-03-02	Female	
4		3	Sauer-Breitenberg	2024-01-12	2023-04-05	Female	

Total rows: 501 | Query complete 00:00:00.085 | CRLF | Ln 19, Col 13

10. Modify the appropriate date column in the database so that when a new record is inserted, the date is automatically set to the current date.

The screenshot shows the pgAdmin 4 interface. The SQL editor contains the following query:

```
6     price,  
7     ticket_discount  
8 )  
9  
10 VALUES (  
11     1002,  
12     2,  
13     'Online',  
14     CURRENT_DATE,  
15     12000,  
16     5  
17 )  
18  
19 RETURNING booking_id, created_at, booking_platform;
```

The Data Output tab shows the results of the query:

	booking_id [PK] integer	created_at date	booking_platform character varying (50)
1	1002	2026-10-07	Online

Total rows: 1 | Query complete 00:00:00.054 | CRLF | Ln 17, Col 52

Successfully run. Total